COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.5 submission template

Questions

7. A standard pack of cards has 52 cards. They are divided into four *suits*: the hearts, diamonds, clubs, and spades. There are 13 cards in each suit, labelled 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King, and Ace. This label is called the *rank* of the card. A *hand* is a set of five cards chosen from the pack.

   A hand of cards is a *two-pair* if it contains two cards which have the same rank as each other, another two cards which have the same rank as each other (but a different rank from the first pair), and a fifth card which has a different rank from all the others.

   How many hands are two-pairs? Explain your answer.

   Write your solution in this box:

14. Let A be a set satisfying $|A| = 6$. Let B be the set $\{x, y\}$. Explain why the number of onto functions from A to be B is $2^6 - 2$.

Write your solution in this box: